Automated Data Analysis Executive Report

Objective: predict diagnosis and explain key factors

1. Validation Confidence Score: 99%

- Sample size adequacy verified: 569 total instances (n=114 test split).
- Overfitting check passed: generalisation gap is within nominal bound (0.7% <= 15%).
- Empirical validation: 4 of 4 hypothesis tests passed statistical significance ($p < 0.05$).

2. Predictive Modeling (LogisticRegression)

Accuracy: 0.982 | F1 Score: 0.982